PHILIPPINE ARMY MANUAL



COMMAND AND CONTROL COMMUNICATIONS, AND CYBER SYSTEMS

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Proponent: AC of S for C4S, G6, PA

PHILIPPINE ARMY MANUAL 6-00

COMMAND AND CONTROL COMMUNICATIONS, AND CYBER SYSTEMS

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Preface

1. This document describes the broad framework to guide the management and employment of Command and Control Communications, and Cyber Systems (C4S). It is based largely on PAM 11-00 (Communications, Electronics, and Information Systems Manual) published in 2009. Aligned with PAM 3-00 (Army Operations), this paper strives to craft an appropriate doctrine for the Philippine Army, specifically the role of C4S in army mission, infrastructures, and the Signal Corps which composes the personnel resources of the system. Other references include the local doctrines and manuals of the AFP and the Philippine Army, and the Army Doctrine Reference Publication (ADRP) 6-0 Mission Command published in 2012 by the US Headquarters, Department of the Army.

2. C4S is founded on the idea that having continuous, uninterrupted flow and processing of information in support of warrior planning, decision, and execution, by making information accessible with its qualities; interoperable, flexible, responsive, mobile, disciplined, survivable, and sustainable. Though this manual, understands the C4S principles for joint and multinational operations that bring together diverse military organizations operating as one force with robust and secure infrastructure. Some of the things it includes are radios, wires, computers, data machines and other means used for the electrical, electro-magnetic and visual transmission, reception, storage and display of information or messages in a clear audio, video or by cryptographic means; all ground electronic devices and system for the control and tracking of vehicles, electronic weather equipment, electronic countermeasure devices; and related electronics systems and equipment. This document is also based on the current capabilities and the future competencies of PA.

3. Since its purpose is to provide Army personnel (officers, EP, and civilian employees) of all levels and in all units of the AFP policies and guidelines for the management of Command and Control Communications, and Cyber Systems (C4S), performance of their duties, functions, and responsibilities.

4. The proponent of this manual is the Office of the AC of S for C4S, G6, PA. However, the Doctrine Center, TRADOC, PA is responsible in the review and updating of this publication. Send comments and recommendations to the Commander, Training and Doctrine Command, Philippine Army, Camp O'Donnell, Sta. Lucia, Capas, Tarlac, Attention: Doctrine Center Ft Bonifacio local landline 4971 or email ddddchtradoc@army.mil.ph.

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Chapter I
Introduction

Command and Control Communications, and Cyber Systems (C4S) is the organization of equipment, methods, procedures, and personnel to accomplish specific information conveyance and processing functions.

0101. C4S is the heart of the technology management employed for the command, control, and administration of the Philippine Army forces. The employment of C4S infrastructure of the Philippine Army possesses key information capability and superiority. As such, C4S is vital to planning, conducting, and sustaining Army operations in logistics and intelligence functions in support of mission command.

The C4S Environment

0102. Command and Control Communications, and Cyber Systems (C4S) has evolved through the years. The signal functions have advanced from complex manual systems to automated systems. This evolution revolutionized the way nations conduct war, and reinvented commanders' command and control of their forces to achieve military victory.

0103. In the past, communications systems were unsophisticated and non-technical. The means of communications consisted chiefly of messengers and mechanically generated signals. The most important and effective means at that time was the commander speaking directly to his subordinates. Centralized management of communications systems was unnecessary, as well as impossible, due to the limited speed and range of communication capabilities.

0104. Communication methods were eventually converted to electronics systems. Elements of standardization were imposed, but the concept of decentralized control remained unchanged. Communications between units were not prioritized as reflected in the limited interface between units as they conducted their missions. The communications network paralleled the command structure. Doctrinal responsibility existed from higher to lower units, from right to left in the battlefield. At every echelon, more than one means of communications were provided. Every signal unit was organized with the goal of providing a self-contained, dependable, flexible, secure, and rapid communications capability.

0105. Basic Communication Principles manual has been developed to provide signal personnel the basic knowledge and theoretical background in handling communications through electronic equipment. The Army Signal Regiment (P) has been assigned as the primary unit of the Philippine Army, providing training on basic electronics to guide signal personnel. This manual may also be used as a reference for personnel taking up career and specialization courses.

0106. The introduction of electronics to communications has changed the battlefield landscape. Time turns out to be the most important factor as distance becomes less

significant. In conjunction with the development in mobility and firepower technology, units now move quicker and shoot more accurately. Thus, the commanders need to be able to exercise pinpoint control. New administrative and logistics systems require the transmission of large amount of data to support the commander. Subscribers now communicate throughout the width and depth of the battlefield. The need for dependability, flexibility, speed, security, and volume capacity increases as units become more mobile and interdependent.

0107. Highly sophisticated communication electronics (C-E) equipment is required to meet the current demands of the Army. To provide voice, teletypewriter, facsimile or data communications across the battlefield, high capacity trunk systems, tactical automatic voice; and data transmission systems are being fielded. These must be integrated at every level to maximize their capabilities and meet the total communication requirements. The commander no longer operates his own communications without concern for systems integration.

0108. C4S technical manual is formulated as a reference material providing technical information and fundamentals of C4S. The use of electronics equipment through experimentation would be vital in enhancing signal personnel's knowledge in the area of electronic communications.

0109. In the advent of emerging information technology, our country's economy is becoming highly dependent on computers. Even the country's defense systems rely on technology in maintaining national security. These are interconnected with the servers, routers, switches and fiber optic cables. This computer-generated automation extends its domain to the cyberspace.

0110. System complexity requires a growing need for automation. This is one of the reasons why a knowledge-based system (KBS) was created. KBS utilizes a set of specialized code language in order to reason and perform limited intelligence tasks. It is made possible by knowledge acquisition through interviews, questionnaires, record reviews, and observation to gain factual and explicit knowledge. KBS is capable of providing expertise to less experienced personnel, avoids delays and provides expertise in locations where it is not available.

0111. C4S has become an important component of the total combat power of an army. Today, warfare is largely dependent on information superiority as information is now both a powerful weapon and target. It is sensed and detected electronically, transported by appropriate communications systems, and processed by computers. Protecting the information held by computers is vital, whilst attacking the opponent's computer systems can produce a significant advantage in the application of force.

0112. These facets have increased the vulnerability of computer networks to cyber-attack. The domain of cyberspace has become a global security concern which, if not assessed properly, will pose threats and could paralyze major critical infrastructures. If not addressed accordingly, threats such as hacks, espionage, and cybercrimes will soon emerge with minimal or no appropriate security measures. For this purpose, Cyber Security manual has been articulated to provide all members of the Philippine Army a basic knowledge on planning, guiding principles, and operational priorities as

well as determining appropriate controls to protect the organization's C4S against cybersecurity threat.

Signal and C4S

Historical Background

0113. The Signal Corps was activated by virtue of the National Defense Act of 1936. Charged with the task of establishing signal communications in the ten military districts of the country, signal officers and personnel were given training and were subsequently assigned to the reserve divisions that were mobilized in 1941.

0114. Due to its relatively new inception, the Signal Corps was the least equipped Army unit. Among the first equipment used were obsolete radios and telephone sets salvaged from the depot at Fort William McKinley (presently Fort Bonifacio). Regardless, the Signal Corps was able to provide crucial service during the 2nd World War. In the battle of Bataan, signal personnel were seen rushing in the midst of the battleground; repairing lines and damaged wires. This eventually led to the establishment of communication that precipitated the conquest of Bataan and Corregidor by the American Liberation Forces led by General Douglas McArthur in early 1945.

0115. In 1947, there was a need to expand and improve the military's signal capabilities. Thus, the Army established Signal Corps school buildings and facilities, stockpiled signal equipment from the American Forces in the Western Pacific, and activated the Signal Branch of the Philippine Ground Force. Additionally, manual telephone systems were replaced by XYPABX systems and high-frequency radio circuits by multi-channel very high-frequency radio systems.

0116. On March 1965, the Signal Service Group was renamed as Communications-Electronics Group (COMMEL) and was tasked to provide, operate, and maintain the communications and electronics facilities on which the AFP Chief of Staff exercised control. The COMMEL Group was then re-organized into the Communications-Electronics Service on 13 November 1978, as a result of its expanding roles and activities.

0117. In keeping pace with the computer age and advances in information technology, the office of the Chief COMMEL was elevated to G-staff status as the Office of the Assistant Chief of Staff for Communications, Electronics, and Information Systems, G11 on 01 May 1996 by authority of HPA General Orders Nr 701.

0118. On 24 January 2005, by virtue of HPA General Orders Nr 40, the Office of the Assistant Chief of Staff for Comptrollership, G6 was deactivated following the re-designation of the Office of the Assistant Chief of Staff for Communications, Electronics, and Information Systems from G11 to G6 effective 01 February 2005 since "6" is more widely used by most foreign armed forces in designating their CEIS staff.

0119. As PA progressed in its organization, the demand for better command and control tools became higher as needed at all levels of Command, from HPA down to the battalion. The commission of a digital Private Automatic Branch Exchange (PABX) at the new HPA Communications Center last 10 October 2001 and the acquisition of Harris Manpack and Handheld radios in 2004 paved way for the Philippine Army to pursue the institutionalization of its entry to the digital age. From this period, the security of the communications facilities of PA became more robust. It also led towards for the modernization Information and Communications Technology (ICT) capabilities of PA-wide units.

0120. The introduction of these new radios to PA inventory meant development of personnel skills to adjust to the operation, use, repair, and maintenance of the newly acquired equipment. Relatedly, to ensure the integration, development, and proper management of CEIS, the Office of the Assistant Chief of Staff for CEIS, G6 of Major PA Subordinate Units (PAMUs) was created on 21 March 2005 for the effective command, control and administration of the unit. This merged the functions of Offices of the Division Signal Officer (DSO), the Signal Officer, and Information Systems Officer (ISO).

Signal

0121. Signal is a means of transmitting information with corresponding pre-arranged system or codes in cases where direct verbal or written statement is unnecessary, undesirable, or impractical. It is the collective and integrated use of Command and Control Communications, and Cyber Systems (C4S) to support warfighting capabilities across all levels of command. Signal support operations are highly essential for the successful execution of army operations.

C4S

0122. C4S is vital for command, control planning and guidance, and in the dissemination of information. It also performs logistics, intelligence, and operation functions. C4S serves as a central system that integrates all aspects of operations and allows the effective and efficient provision of command and control communications.

0123. Information superiority is a key advantage in modern times. For the military, it achieves operational advantage, which allows increased situational awareness, better decision-making, and connectivity in the battle sphere. The role of C4S is to ensure the delivery of robust and flexible solutions to situational disadvantages in military operations.



Figure 1.1 – Signal Functions in the Philippine Army

The Signal Functions in the Philippine Army

0124. As shown in Figure 1.1, the following are the basic signal functions in the Philippine Army:

a. **Sensing and Detection**. These are the signal functions of collecting information particularly on "beyond visual range" (BVR) targets. These also deal with the tracking and identification of enemy or friendly forces' location.

b. **Provide Infrastructure for the Processing of Information**. With the aid of computer hardware and software, the signal automates processing of information for timely decision-making and responsive intelligence.

c. **Protection of Information**. In the current cyber warfare, information is becoming the object of enemy activities; thus, it is both a weapon and a target that must be protected.

Signal Support Operations

0125. Signal organizations exist at every echelon of the Philippine Army. Their mission is to support the commanders in performing command and control, administration, and intelligence functions to accomplish their mission.

0126. As the standard guiding doctrine, a Signal Support Operations manual is formulated. It lays out common understanding, organization, and deployment of PA signal assets to support the C4S requirement of the Army.

Signal and C4S

0127. Signal is the action that supports Army operations. On the other hand, C4S consists of capabilities, enabling signal to support Army operations.

Chapter II
C4S Philosophy, Principles, and Concepts

0201. The following fundamental principles shall apply in the formulation and implementation of policies, plans, programs, and projects on C4S:

a. **Command Responsibility** – C4S formulation and implementation are the responsibilities of command. Commanders in all echelons are responsible for the establishment, operations, and maintenance of C4S infrastructure within their respective units, as well as their efficient interconnection with other units.

b. **Inseparability of Operations and C4S** – Military operations and military C4S are integral to each other. C4S must be provided and maintained to obtain maximum overall effectiveness of the forces consistent with the projected rate of activity and scope of operations.

c. **Advance Provision of C4S Infrastructures** – C4S should be provided in advance of projected operations, whenever practicable.

d. **Need for Common Procedures and Lateral Interconnections** – Functionally similar systems should be interrelated by use of common procedures and lateral inter-connections. Systems planning and procedures must permit transmission of messages from any point in one system to another, without major delay or reprocessing.

e. **Relative Importance of Reliability, Speed, and Security** – C4S infrastructures must be reliable, rapid, and secure. Reliability is paramount under all circumstances. The importance of speed may transcend the need for security, but the relative importance of these two must be weighed in light of current conditions. In addition, C4S must be simple, flexible, and economical, without outweighing reliability, security, and speed.

Operational Concepts

0202. The following operational concepts shall apply in the establishment, operations, and maintenance of C4S infrastructures:

a. **Survivability** – C4S infrastructures must be designed in such a way that destruction in a portion of a system will not disrupt the entire system.

b. **Flexibility** – A high degree of flexibility is required for C4S infrastructures. The wide dispersion and multi-tasking of units require a flexible C4S infrastructure that can readily adapt to changing situations and diversified type of operations.

c. **Economy of Force Principle** – This principle leads towards the most efficient use of personnel and materiel available to every commander. Along this line,

C4S is integrated among commands and services consistent with reliability and flexibility. For example, part of the needs of a certain commander may be provided by allocation of channels in the system of a higher commander, or a commander may ask a subordinate commander to provide certain communications channel for use of the higher commander's systems or for another subordinate commander.

d. **Uni-Service Overall Facilities –** The principle of economy of force makes it advisable that only one service maintains overall C4S facilities between two geographic points whenever practical and consistent with reliability. These service facilities would also provide facilities communications channels for the use of other services. When a single facility has been agreed upon to serve more than one command or service, it should normally be under the operational, maintenance, and security control of a single command as determined by a competent authority. This command will normally have primary interest and responsibility in the use of the facility.

e. **Common Use of Channels –** The common use of communications channels by more than one command or service will be a normal practice; thus, the exclusive allocation of channels is not advisable. Standards of service, as established by proper authority, must be maintained at such a facility.

f. **Inter-Operability and Standardization** – The effective joint communications and integration of C4S facilities must be standardized and interoperable. Plans for standardization and interoperability must ensure that the essential requirements of all services are accommodated.

g. **Restricted Use** – Military C4S facilities are provided primarily for commanders, chief of offices, and their staff, in support of mission accomplishment. Excessive or extravagant use of facilities is discouraged as it will not only waste limited resources but also nurture habits and expectations that are counter-productive, especially in times of emergencies. Military C4S facilities will only be provided to those authorized by competent authority.

Appropriate Technology

0203. In the establishment of C4S facilities, the latest technology is not necessarily the best choice since other factors such as viability, suitability, technology risks, manpower limitations, and others, may point to another resource as the more appropriate choice.

PA C4S Philosophy

0204. The PA C4S Philosophy is a reliable, responsive, secure, and cost-effective C4S infrastructure that enables the Army to complete its mission. It involves the inter-relationship of the C4S sub-systems such as the communications system, electronics system, information system, and cyber system.

0205. It is also known as the voice of command as it provides Philippine Army with the necessary information that flows from the Commanding General down to the lower commands and units.

0206. This shall provide the unifying concept and underlying rationale in the formulation and implementation of policies, plans, programs, and projects involving the C4S infrastructure of the Philippine Army.

0207. C4S is an integral part of the PA force structure and weapons systems. It provides linkages and the information management tools for a safe, accurate, and timely sensing, detection, and processing of information. Therefore, the Philippine Army shall develop and maintain an effective C4S infrastructure for the command and control, intelligence, and administration requirements based on accepted principles, operational concepts, and appropriate technology.

Conceptual Framework.

0208. A systems approach (input-process-output) as shown in Figure 2.1, is used to portray the relationship between the C4S philosophy and other inputs (requirements and resources) with the formulation and implementation processes (policies, plans, programs, and projects) towards the desired output of a reliable, responsive, secure, and cost-effective C4S infrastructure. The philosophy serves as an input variable and at the same time a harmonizing agent in matching the requirements (for command and control, intelligence, and administration) with the available resources (personnel, facilities, materiel, funds, and time) subject to the formulation and implementation process. A feedback loop is likewise shown to emphasize the need for assessing the performance and acceptability of the C4S infrastructure as a basis for modifying any of the identified input factors and processes. (Figure 2.1)

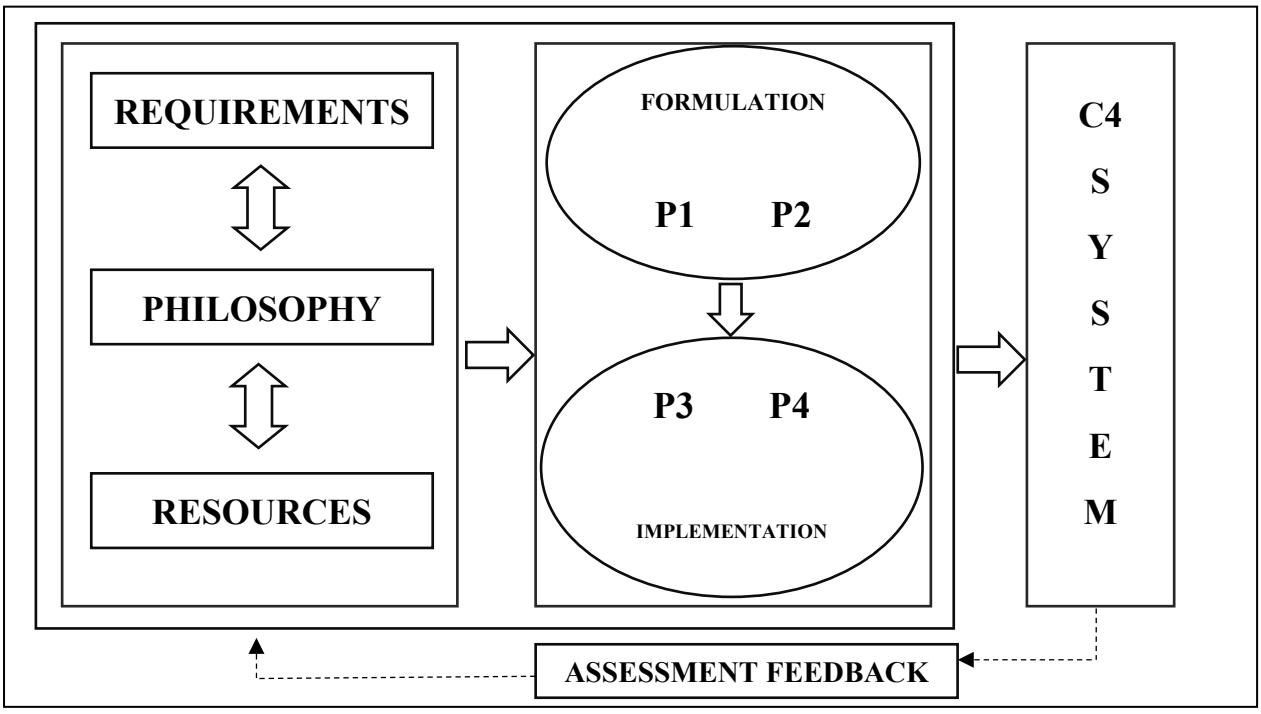


Figure 2.1 – C4S Philosophy Operational Paradigm

Role of PA C4S

0209. C4S enables communications to flow from the Commanding General of the Philippine Army down to the lower levels of command that includes front line units and elements.

0210. The high level of technology necessitates accuracy and timeliness. Access to information becomes necessary in planning, directing, controlling, and coordinating elements of forces. The C4S support infrastructure serves as the linkage and information management tool for sensing, detection, processing, and exchange of information, besides coordination of control, combat, as well as for other support related purposes.

0211. The C4S support infrastructure provides for the following command and control requirements:

a. **Common Operational Picture**. The C4S infrastructure provides the required situational display, messaging, database access, and other functionalities needed to provide common, timely, and accurate viewpoint of the current operational situation. A common operational picture is a view of the situation at the lowest tactical element or unit which is similar to the intervening headquarters up to the Army Operations Center and the General Headquarters.

b. **Collaborative Planning for Military Operations**. The C4S infrastructure allows key operational planners to collaborate electronically at their primary locations in the preparation, coordination, and distribution of operational plans.

c. **Centralized Command and Decentralized Execution**. The C4S infrastructure furnishes vital information to serve as basis for immediate and effective decision-making. It provides the required capabilities to rapidly distribute directives and guidance from senior military leaders to forces responsible for the execution of mission within the operational area.

d. **Integration of Intelligence and Operations**. The C4S infrastructure provides intelligence and operations staff with the capability to promptly receive, analyze, and exchange the latest intelligence needed for operational planning and decision-making.

e. **Apportionment and Reapportionment of Resources**. The C4S infrastructure provides the commander the capability to immediately assemble and analyze information in the current situation, identify operational requirements, and verify the availability and location of required resources. It also provides the capability to efficiently issue directives related to the apportionment and reapportionment of available resources to meet the changing situational needs and demands.

f. **Command and Control Information Push and Pull**. The C4S infrastructure provides commanders in all levels the capability to deliver

information to both superiors and subordinates (information push), and access reliable information sources for specific information needed in decision making (information pull).

g. **Security** – The C4S infrastructure ensures that the information goes only to those who need to receive it. It must be able to withstand cyberspace attacks.

C4S Response to Humanitarian Assistance and Disaster Response (HADR) Operations

0212. A sudden disastrous event produces great material damage, loss, and distress. Disasters include human-induced disasters such as civil strife, terrorism, conflagration, and epidemic outbreak, to name a few.

0213. The C4S Support Plan for HADR shall be accomplished in four (4) phases:

a. **Disaster Prevention and Mitigation Phase** – involves the formulation and establishment of PA C4S policies, plans, projects, and programs aimed at developing effective C4S capabilities for, but not limited to, electronic warfare (EW), computer network operations (CNO), information assurance (IA), communications security (COMSEC), and network security (NETSEC) to support HADRO. This includes the synchronization of all radios; conduct of communications exercises; training and re-training of personnel on the operation and maintenance of C4S equipment; preparation of CEOI, communications plan, standardized reporting procedures, and HADR communications network management structure, among others.

b. **Disaster Preparedness –** are activities that limit the impact of disaster by structuring the response, and providing quick and effective actions after the catastrophe. It addresses actions in both the pre-disaster and post-disaster phases. It includes assessing vulnerability; conducting and introducing disaster research and related technology (e.g. satellite warning system, tsunami or lahar monitoring stations); supporting organizations and institutions that plan and implement disaster response programs; maintaining regional disaster stockpiles; and training. Disaster preparedness involves finalization and updating of communications network plans and projects; dissemination of hotlines; and issuance of warning orders to ensure interoperability among military troops and other responding agencies, international organizations and foreign allies during HADRO. Action plans that address concerns are as follows: (a) communication plans, (b) emergency response teams, (c) emergency warning methods, (d) shelters, evacuation plans, (e) resources inventory building, (f) development of trained volunteers, and (g) casualty prediction.

c. **Disaster Response -** involves the mobilization of emergency teams, core emergency services, specialist teams, and first responders in area. Search and rescue efforts run on the principle of unified command, mutual aid, and addressing immediate needs. Disaster response involves effective deployment and monitoring of C4S facilities and resources to ensure reliable, robust, and redundant communications in support of HADRO requirements for information exchange. Disaster response operations involve concerted efforts by two or more agencies, governmental and/or otherwise, to provide emergency assistance in relief to victims of a disaster/calamity, and in the restoration of essential public activities and facilities. Specific aid and assistance that may be

provided in disaster operation includes: the issuance of medical supplies, equipment, and emergency medical treatment; food, water, and shelter; rescue and firefighting services; police protection; route clearance and traffic control; prevention of panic; communications; and restoration of facilities.

d. **Disaster Rehabilitation and Recovery -** ensures that the deployment of C4S facilities and C4S resources provide operating troops in affected communities with alternative means of communication while restoration of communications infrastructures and services are on-going. Disaster recovery also enables the restoration of affected area back to its normal and conducive state by rebuilding, employing, and repairing damaged facilities. It seeks for a "window of opportunity" in implementing measures of mitigation.

e. **Disaster Mitigation -** concentrates on reducing the harmful effects of disaster. It focuses on long-term measures, and is part of recovery process. Mitigation actions maybe structural (use technology) or non-structural (national legislation). To be most cost-effective, it necessitates identification of risks.

C4S Support to Mission Command

0214. In order to effectively implement mission command, each commander needs to establish a system composed of networks, information systems, processes, procedures, facilities and equipment to enable the conduct of operations.

0215. Through this, it can: a) support the commander's decision making; b) collect, create, and maintain relevant information and prepare knowledge products to support the commanders' and leaders' understanding and visualization; and c) establish the means by which commanders and leaders communicate, collaborate, and facilitate the functioning of teams.

0216. Networks are set of things that are interconnected – established to enable commanders to exchange ideas, build teams, and promote unity of effort. It is made possible by information systems. Commanders are aware that they face risks such as networks being degraded during operations. Thus, they develop ways on how to lessen the impact of such risks.

0217. Commanders arrange facilities and equipment (vehicles, signal equipment, lighting) that allows components of the mission command system to work successfully.

Chapter III
C4S Operational Process

C4S as a Functional Area and Relationship

0301. **C4S**. Command and Control Communications, and Cyber Systems (C4S) is classified as a functional area or major activity of the Philippine Army. It is the activity involving the management of C4S infrastructures.

0302. **C4S Infrastructure is a resource**. C4S is composed of the linkages and information management tools for the commanders in decision-making. It enables him to plan, direct, control, and coordinate the elements of his forces to accomplish his unit's mission.

0303. **AC of S for C4S, G6**. The Assistant Chief of Staff for Command and Control Communications, and Cyber Systems (C4S), OG6 is responsible to the Commanding General, Philippine Army in the formulation of policies for the management of C4S and its corresponding resources for the Philippine Army.

0304. **The Signal Corps, Philippine Army**. The Signal Corps is classified as a primary AFPOS in the Philippine Army. It provides the personnel resources for the establishment, operations, maintenance, and training of the various C4S infrastructures. It is composed of personnel who were trained and/or actually engaged in the signal activities of the Philippine Army.

0305. **Signal Personnel**. This refers to Army personnel qualified to perform signal functions as defined in this manual. Every member is authorized to affix to his name the letters "SC" and wear the official AFP signal corps badge.

0306. **Chief, Army Signal Corps**. The officer designated as chief of the Army Signal Corps is the commander, ASR (P). Part of his functions is the career management of signal officers and PA enlisted personnel.

0307. **Functional Relationship.** The Assistant Chief of Staff for Command and Control Communications, and Cyber Systems (AC of S for C4S) shall have staff supervision over ASR (P), PA. The C4S guidelines on the deployment of ASR (P), PA took effect dated 29 April 2011.

0308. **The Subdivision of Signal Corps AFPOS.** The following is the revised Philippine Army military occupation specialties (MOS) that fall into the signal field (Command and Control Communications, and Cyber Systems-C4S). These work together to make all units able to communicate with each other.

Table 3.1 Subdivision of Signal Corps AFPOS

Signal Areas	Military Occupation Specialties (MOS)	Code
Signal, General		
Signal, General	Signal System & Materiel Integrator	**22A**
	Signal Educator	**22B**
	Signal Planner	**22C**
	Signal Support Systems Specialist	**22D**
Command and Control Communications System (C3)		
Communications System	Radio Operator/Maintainer	**22E**
	Cable Systems Installer/Maintainer	**22F**
	Network Switching Systems Operator/Maintainer	**22G**
	Nodal Network Systems Operator/Maintainer	**22H**
	Microwave Systems Operator/Maintainer	**22I**
	Multichannel Transmission Systems Operator/Maintainer	**22J**
	Satellite Communication Systems Operator/Maintainer	**22K**
	Radio Programmer/Technician	**22L**
Electronics System	Electronics System Technician	**22M**
	Audio Video System Operator/Maintainer	**22N**
	Electronic Warfare Operator/Maintainer	**22O**
	Electromagnetic Spectrum Manager	**22P**
Information System	Information Technology Technician	**22Q**
	Information Technology Specialist	**22R**
	Network Systems Operator/Maintainer	**22S**
	System/Network Administrator	**22T**
	Software Developer/Programmer	**22U**
	Storage and Data Specialist	**22V**
Cyber System		
Cyber System	Cyber Operations Specialist	**22W**
	Cyber Network Defender	**22X**
	Network Security Operator/Maintainer	**22Y**
	Computer Incident Responder	**22Z**

Required C4S Infrastructure at the Three Command Levels

0309. The PA C4S architecture conceptual framework develops the analysis of the required C4S infrastructures at the three command levels: strategic, operational and tactical). Likewise, it evolves from the translation of these requirements into specific programs and projects which are functionally integrated and synthesized into the three

C4S generic systems, namely: communications systems, electronics systems, and information systems.

0310. Equally, the C4S Network Infrastructure manual is formulated to assess and to focus on the three command levels (strategic, operational and tactical) and on how the C4S generic systems can be integrated to these levels. It is a doctrinal basis to describe the responsibilities, relationships, procedures, capabilities, constraints, and planning considerations in the installations, operations, and maintenance of C4S infrastructure.

Strategic C4S Infrastructures

0311. These are C4S capabilities employed at the international and national levels, utilizing predominantly fixed type, in support of the national command authorities, GHQ-AFP, the headquarters of the major services, and other agencies. These strategic players must provide command and control linkages to the operational C4S and tactical C4S players. Strategic C4S infrastructures also include special purpose and service peculiar systems such as the PAF Air Defense Surveillance System and the PN Coast Watch System, respectively centrally operated on a nationwide scale by the Philippine Air Force and the Philippine Navy. These systems are normally established by GHQ-AFP.

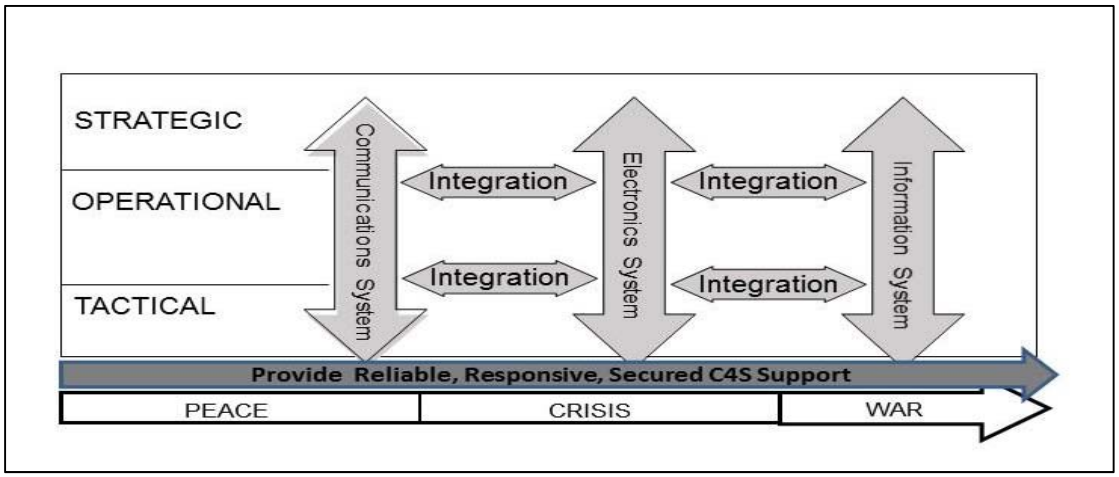


Figure 3.1 - PA C4S Conceptual Framework

Operational C4s Infrastructures

0312. These are C4S capabilities employed at the sub-national level utilizing a mix of fixed and mobile facilities to support of area or unified command or joint task force headquarters. Command and control linkages are provided to the strategic C4S and tactical C4S.

Tactical C4S Infrastructures

0313. These are C4S capabilities employed at the field operating level utilizing predominantly mobile or transportable and self-contained facilities in support of deployed operating forces.

Chapter IV

C4S Sub-Systems

0401. The PA C4S is functionally structured into four interdependent systems as follows: communications systems, electronics systems, information systems, and cyber systems. Each system is further sub-divided into sub-systems, depending on the capabilities they should provide to the three command levels.

Communication Systems

0402. Through communication systems, data are transmitted to the commanders and staff for information dissemination. Communication systems play an important role in providing adequate information to the commander and staff, in setting procedures and in making decisions to sustain shared understanding and to strengthen connections within the command. The communications systems will serve as the PA telecommunications network, which shall provide the linkages from HPA to its subordinate units; from HPA to higher headquarters including the national command authorities; and from HPA to other agencies of government and the private sectors. The communications infrastructure shall support the transfer of information whether in the form of voice, data, image or video. Communications systems are broadly categorized into fixed, mobile, and special systems, depending on how they are used.

0403. **Fixed Communication Systems.** These are the permanently installed communications infrastructures providing telecommunications services within and among PA camps and bases throughout the country. The PA fixed communications systems, like any other fixed communications systems, may be broken down into the following major components, namely:

a. **Transmission Facilities** – These refer to the information superhighway between switches, normally provided to by GHQ-AFP operated communications systems, such as the AFP fixed communications systems (AFP FCS).

b. **Switching Facilities** – These provide the routing of information from the source to the desired destination. In the Philippine Army, these include the switching facility at HPA and the switching facilities of different army camps and bases nationwide. In conjunction with the transmission facilities, these switching facilities form into a network of switches for the delivery of the needed voice, data, image and video services.

c. **Cable Facilities** – These refer to the distribution lines interconnecting the switching and terminal facilities.

d. **Terminal Facilities** – These are devices such as telephone, computer systems, television sets, and other multi-media devices that interface with the user.

e. **Support Facilities** – These include buildings, power generators, commercial power and water supply facilities, environmental control and other ancillary

equipment, which are required to support the operations and maintenance of the fixed communications infrastructures.

0404. **Mobile Communications Systems**. The PA mobile communications systems cover those used in support of army operations when units and troops are on the move or are temporarily halted. Because of this nature of employment, the PA mobile communications systems are normally portable or transportable intended to support field operations. It should support rapid deployment and should be characterized by flexibility, simplicity and ruggedness. Mobile communications equipment shall be specified so that they will not hamper the capability of the individual soldier or unit to move swiftly and to shoot accurately.

0405. **Special Purpose Communications Systems.** These systems are envisioned to enhance the capability for command and control in areas where, due to terrain limitations and peculiar operational environment could not be provide regular links to the AFP FCS. Projects in this category include the very small aperture terminal (VSAT) satellite systems, mobile satellite communications systems, commercial cellular telephone system, and others that may be appropriate in any particular operational environment.

Communications Connectivity between the Different Command Levels

0406. Considering that communications systems shall provide the linkages from HPA, the following define the appropriateness of each type of communications systems to be used as connectivity means between the different command levels.

a. **Within Strategic Level Headquarters** – The primary connectivity among strategic level headquarters shall be provided by the AFP Fixed Communications System established by GHQ-AFP and is operated and maintained by CEISSAFP. Appropriate HF/SSB radios providing the same level and type of capability and services as the AFP FCS shall be established as secondary means of connectivity. Commercial landline services may be established as tertiary means of connectivity. Commercial mobile communications systems such as cellular phones shall be subscribed to provide the means of communications connectivity while on the move and while the AFP has not yet established its appropriate mobile communications systems.

b. **From Strategic Level Headquarters to Operational Level Headquarters** – The primary connectivity at this level shall be the AFP FCS, as defined in the preceding paragraph. The Philippine Army shall provide drop and insert connectivity from appropriate AFP FCS breakouts to connect major PA camps. Appropriate HF/SSB systems providing the same level of capability and services, as with the AFP FCS, shall be established as the secondary means of connectivity. Commercial landline services may be established as the tertiary means of connectivity, subject to available resources. Moreover, commercial mobile communications systems shall be used as appropriate and if there are provisions to pay for such services.

c. **From Operational Level Headquarters to Tactical Level Headquarters** – The primary connectivity at this level shall be the AFP FCS as previously defined, when technically feasible. Otherwise, appropriate HF/SSB systems providing the same level of capability and services as with the AFP FCS shall be used together with appropriate VHF/FM or UHF/FM systems. In this context, the PA shall endeavor to integrate the VHF/FM or UHF/FM systems through the establishment of repeater systems at strategic areas. For these purposes, commercial landline or mobile communications systems may be used as appropriate.

d. **From Tactical Level Headquarters to Operating Units' Headquarters** – As in the preceding paragraph, the primary connectivity at this level shall be the AFP FCS when technically feasible. Otherwise, HF/SSB system shall be used. Appropriate VHF/FM or UHF/FM systems as in the previous paragraph shall be established. Likewise, commercial landline or mobile communications systems may be used as appropriate.

e. **From Operating Unit Headquarters to Unit Elements** – Appropriate mobile communications systems shall be used at this level. Standardization of mobile communications equipment shall be observed to enhance inter-operability.

Electronics System

0407. Electronics systems are those which functions are related to, but not primarily, communications. These systems include, though not limited to, any of the following intelligence, battlefield surveillance, land navigations, electronic warfare (EW), public address, night fighting systems, visual presentations, target acquisitions, combat camera, and public affairs.

0408. Through such systems, the amount of automation and ease of use increases , enabling the Army to develop sensing and countermeasures. These often include computer modelling and simulation. This is to protect personnel, facilities, and equipment from any influence by friendly or enemy forces that degrade, neutralize, or destroy combat capability.

0409. On the planning and conduct of electronic warfare operations, SIMBn of ASR (P) has put together a doctrine on EW to establish a better and common understanding on the installation and maintenance of C4S equipment, which includes the installation and repair of EW equipment.

0410. EW support empowers the commander to intercept, identify, search, and locate intentional and unintentional radiated electromagnetic energy. Thus, the PA forces can determine the electromagnetic vulnerability of an enemy through carrying this out. As a result, the PA forces can do targeting, planning, and distinguishing threats on future operations.

0411. The components of electronics system are all integrated to deliver up-to-date information and a common operational picture of the area to be safeguarded at tactical, operational and strategic levels of command.

Information Systems

0412. Information systems cover the enhancement of the computer-based process of sourcing, manipulating, and storing data for the purpose of generating information for the Philippine Army utilization. The development of this capability ensures timely and detailed monitoring of a wide array of information, thus, allowing top echelons of command to quickly assess emergency and crises situations and to make the necessary decisions, maximizing the use of automated command and control processes.

0413. The Network Enterprise Technology Center, ASR (P) of the Philippine Army laid out a standard guiding doctrine on C4S in providing Information Technology based services to the major PA units.

0414. **Computer Networks**. The PA computer network defines the depth and the breadth of the physical connectivity of computers within the AFP, both in support of the command and control as well as those of the day-to-day administration. In support of the command and control process, the network identifies at the very least, the link between the HPA to the GHQ-AFP, the NCA and other agencies, besides the connectivity from HPA to the different infantry divisions and other PA-wide support and separate units. These shall be used primarily for the conduct of AFP operations in times of peace, crises and war. In the case of day-to-day computing, the servers and telecommunications linkages may support area common-user requirements for general computing activities. The computer network of the Philippine Army shall be called PANET (Philippine Army Network). The primary hub shall be established at the PA communications center in Fort Bonifacio.

0415. **Information System Software**. As in any large and complex organizations, the PA needs wide array of information in order to accomplish its mission. The following key information systems of the Philippine Army shall be developed:

a. **Strategic Planning**. This process embraces information concerning policymaking strategy formulation, international relations and national development program. Policy-making is the formulation of guidelines that shall govern the planning and programming of PA strategic activities. International relations information refers to the national government policies concerning other countries that have direct or indirect bearing on Army operations and projections. The national development programs shall serve as a reference point in execution of the nation-building role by the Army.

b. **Army operations**. Army operations information agenda include a comprehensive view of combat operations, civil-military operations and organizational development programs for responsive planning and control. Information on combat operation should allow the commanders to see “the battlefield” and to take quick and decisive actions on the matter. Civil military operations (CMO) shall help the Army deal with the populace both in the battle and non-battle areas, thereby avoiding unpopular behavior and actions as well as negative impressions, which may affect the winning momentum of the troops. Information on organizational development programs; on the other hand, shall enable the Army to create a relevant and potent fighting force.

c. **Program management**. Information on program management shall monitor progress and indicate the state or direction of strategic and development efforts of the Army. Program management deals with both short and long range plans and strategic measures for implementation, monitoring and evaluation in line with objectives or desired outcomes.

d. **Resource management**. Resource management information should be concerned with the accurate allocation, fair distribution and wise utilization of resources for effective administration and control. To that end, PA should maximize the use and minimize waste of these scarce resources to support adequately all of its intended missions.

e. **Functional and Staff Administration**. This aspect of information management provides the operational day-to-day need for information of the different functional activities and staff offices, in support of the performance-based and productivity-oriented staff management.

Cyber Systems

0416. It establishes the classification, downgrading, declassification, transmission, transportation, and safeguarding of information. In doing so, it requires protection of the interests of national security, consistent with the dynamic and fast changing technological advancements in achieving acceptable, reliable, resilient and industry-standard levels of security, protecting PA C4S' critical infrastructure.

0417. With the advent of information technology, the country's economy is becoming highly dependent on computers. Thus, the country's defense systems now rely on this in order to maintain national security. Systems are all interconnected with the servers, routers, switches, fiber optic cables, wireless devices, satellites, and other mechanism, which allow critical infrastructures to work immensely to have a functionally stable economy.

0418. There is no doubt that cyberspace has made communication capabilities limitless. However, because of the vast amount of resources available, certain circumstances create critical vulnerabilities which adversaries or enemies continually exploit. With continuous advancement of information technology in the present age, alongside with its opportunities and risks, the Philippine Army adopted cyberspace as the fourth domain of operations. As such, comprehensive strategies and procedures must be performed in order to protect the Philippine cyberspace. This domain must be included in the Army's strategies, policies, doctrines, operational plans and orders to achieve information superiority, and to defend and protect its critical information infrastructures and vital systems from disruption, destruction, manipulation, or exploitation.

0419. By recognizing cyberspace as its domain, the AFP acknowledges that modern battles are waged not only in air, sea, or land but also in the virtual setting. As such, the cyberspace must be considered as one of the factors to be analyzed in the conduct of planning and execution of all AFP operations and capability development, both materiel and non-materiel.

Chapter V
C4S Management

The Development of PA C4S

0501. The passage of Republic Act (RA) 10349 on April 11, 2013, known as the Revised AFP Modernization Act, has given the AFP a fresh mandate for the development of its capabilities. RA 10349 amends the Act No. 7898, which establishes the former and its other purposes policy of the state in particular. It aims to modernize the AFP at a level which effectively and fully performs its constitutional mandate to uphold the sovereignty and to preserve the patrimony of the Philippines. The enactment of the Revised AFP Modernization program enables the AFP to enhance its capability to fulfill its mission of protecting the Filipino people and of performing responsibly on conventional and/or unconventional warfare, disaster relief and rescue operations, besides contributing to economic development and other non-traditional military roles. The Act promulgated five (5) major components, namely; a) force restructuring and organization development; b) capability, materiel and technology development; c) bases/support system development, d) human resource development and; e) doctrine development.

Materiel Management

0502. Management of materiel equipment for communications and electronic system is essential in the performance of signal capability. Focus is given particularly to mission essential equipment, in the absence of which, the mission might be jeopardized or may not be carried out at all.

Capability Upgrade Program (CUP)

0503. The CUP is established for the acquisition of mission-essential equipment for internal security operations (ISO). Its objective is to enhance the readiness level to at least R2 (or partially mission-capable) if not R1 (fully mission-capable).

The PA Command, Control, Communications, Computer, Intelligence, Surveillance, Target Acquisition and Reconnaissance (PA C4ISTAR)

0504. The C4ISTAR system is the integration of command, control, communications, computer, intelligence, surveillance, target acquisition and reconnaissance functions to arm the commander at all levels of command with accurate, timely and correct information directly from the battle ground. A reliable C4ISTAR system will enable the commander to deploy forces and resources in an effective and efficient manner towards mission accomplishment. The PA has been developing its C4ISTAR System as an indispensable part of interconnecting system of information highway. Its sub-components are the following:

a. Command and Control
b. Communications and Computer

c. Intelligence, Surveillance, Target Acquisition and Reconnaissance (ISTAR)

d. C4ISTAR Support

0505. **Command and Control**. The command and control component of C4ISTAR arms the commander to effectively and instantaneously direct all its organic and attached forces in order to accomplish the assigned mission. A decision support tool must be available to integrate all the available information so as to develop the best courses of action to any combat and non-combat situations. The system shall emphasize on situational awareness and attain a Common Operation Picture (COP) at all levels of command. The AOC and the Tactical Operations Center (TOC) of major PA subordinate units must support a full C4ISTAR capability with its various applications, such as: intelligence gathering and dissemination, secured voice and telephony and video teleconferencing. The structural design of the command and control architecture is shown Figure 5.1.

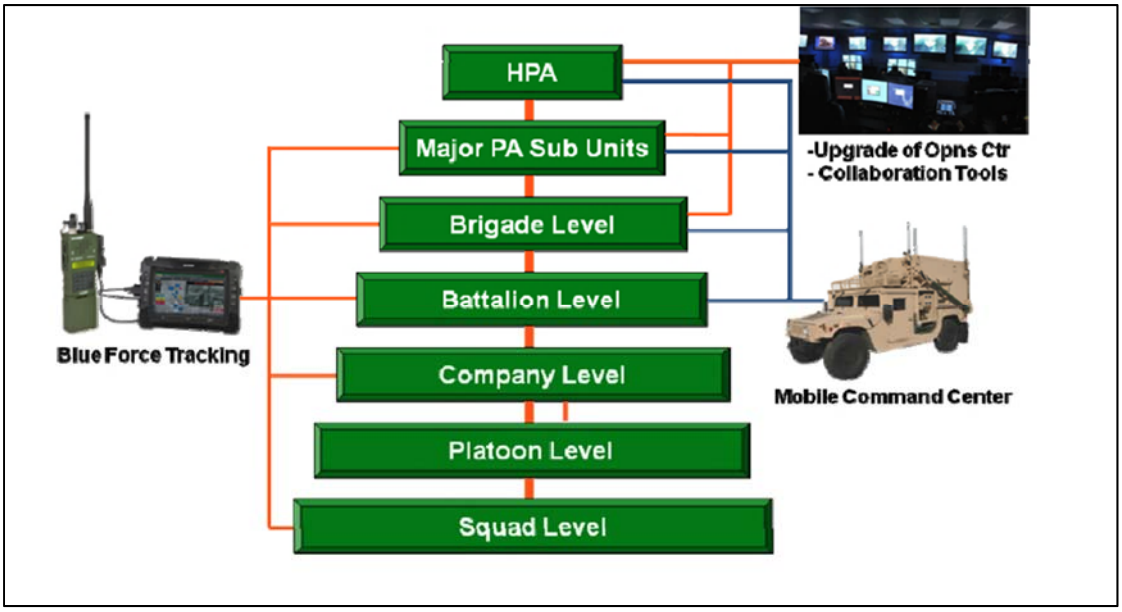


Figure 5.1 - PA Command and Control Network Architecture

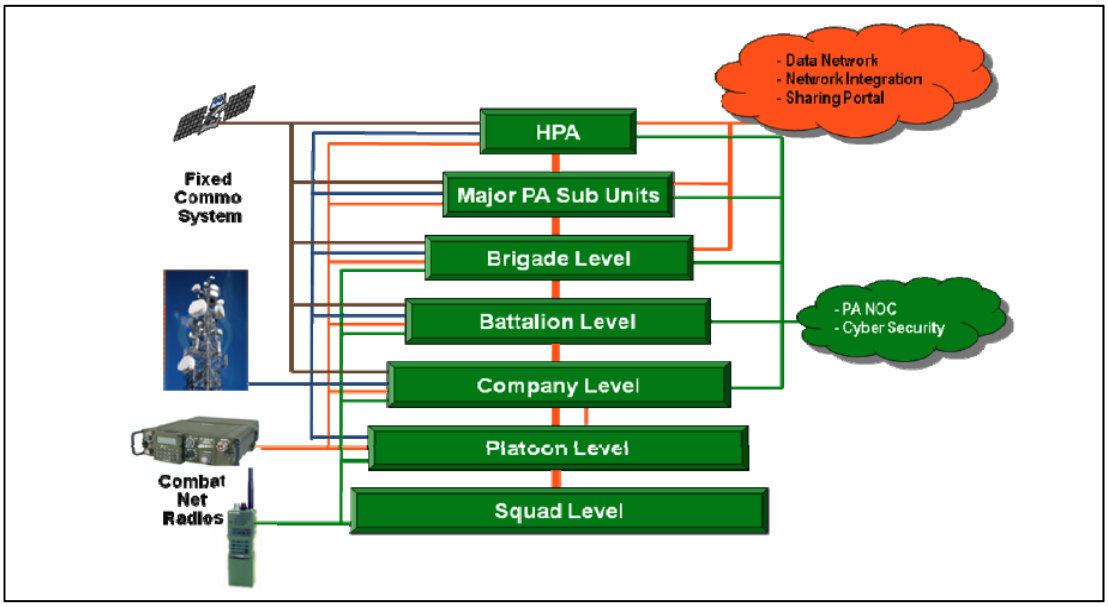


Figure 5.2 - PA Communications and Computer Network Architecture

0506. **Communications and Computer**. This sub-component is aimed to channel the flow of communication throughout the whole organization and to process the influx of gathered information. Doing so shall provide integrated and interdependent communications infrastructure for voice, data and video teleconferencing. To foster interoperability, this shall bridge the vertical gap between the AOC and TOCs of the PAMUs down to the squad level, as well as the horizontal gap army AFP units, PNP, and other government agencies. The network connectivity to be used shall be secured and must be robust for it will be the backbone of the whole system which applies in all operational centers. Figure 5.3 illustrates that AFP C4ISTAR network infrastructure. This network shall observe fully integrated fixed communications system (FCS) which will interconnect all AFP units as follows; tactical communications equipment for the field units; PA network operations center (which will duplicate the existing network operations center at Network Enterprise Technology (NET) Center for the safety of stored data); data network infrastructure for the headquarters of PAMUs; network integration; satellite communications; cyber security system to monitor and counter attacks and threats to the PANET; and an information-sharing portal.

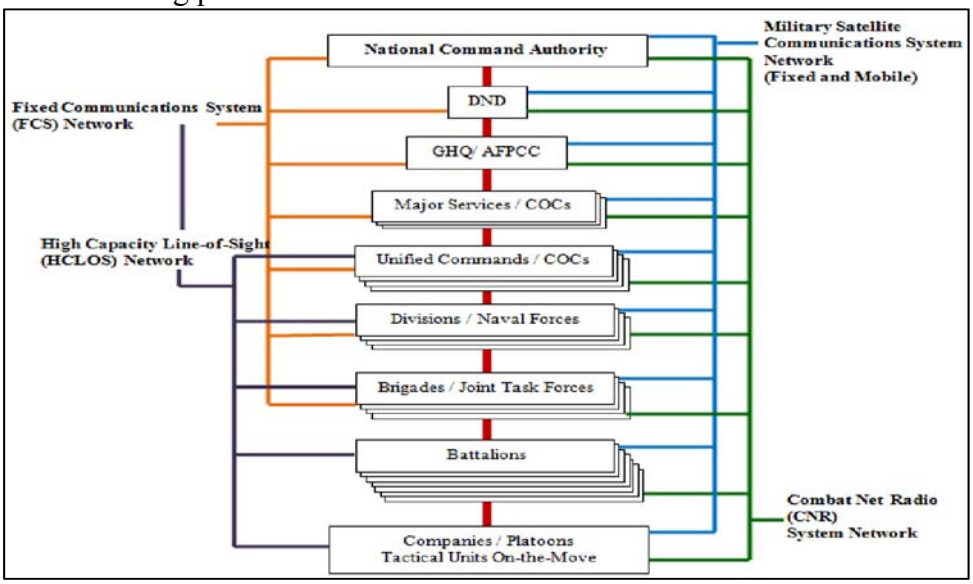


Figure 5.3 – AFP C4ISTAR Network Infrastructure

0507. **Intelligence, Surveillance, Target Acquisition and Reconnaissance** or ISTAR capabilities encompass gathering, communication, consolidation, and processing of information central to leveraging of tactical units that are operating in the field. In general, ISTAR employs a variety of technological capabilities throughout the process, but it underscores the use of unmanned aerial vehicle (UAV) to acquire visual intelligence and produce real time information

0508. In this system, intelligence units do not have the monopoly of data-gathering technologies because appropriate tools shall be shared with combat units which are strategically capable of gathering much needed information in the field. Ultimately, the combat units will be the end-user of intelligence processed from the ISTAR capabilities as they engage the enemy and send accurate data for the forward support.

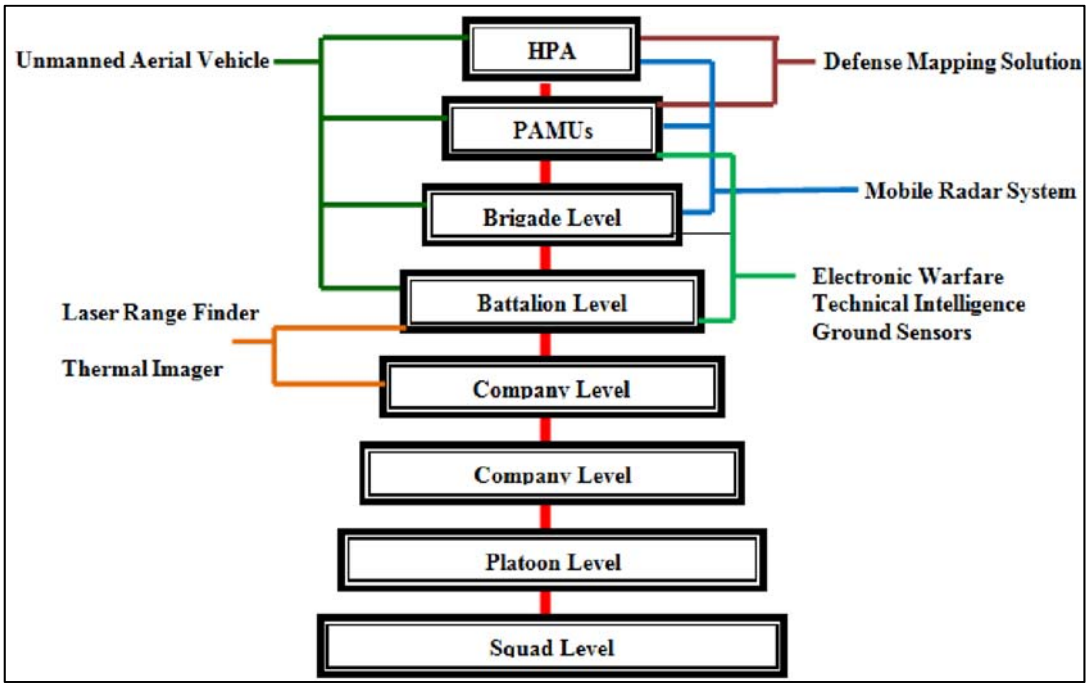


Figure 5.4 - PA ISTAR Architecture

0509. It will also be useful in determining the accurate location of the enemy for a precise targeting of fire support delivery from the artillery. The ISTAR capabilities shall have the following architecture as shown in Figure 5.4.

0510. **C4ISTAR Support**. The proposed PA C4ISTAR System shall have auxiliary sub-systems to ensure continuous and institutionalized service and operation. This will require training and simulation facilities, repair and maintenance facilities, and above all, doctrine development.

PA C4S Management Strategy

0511. The ultimate objective of the management strategy is to provide a C4S infrastructure that is reliable, responsive, secure and cost-effective, taking into considerations the requirements of the Philippine Army, the evolving C4S technology,

the available resources, and the C4S philosophy. The management strategy is embodied on the following thrusts as shown in Figure 5.5:

a. Maintenance of Existing Systems

b. Improvement thru Innovations

c. Development of Appropriate Systems

d. Sustained Strategy Implementation

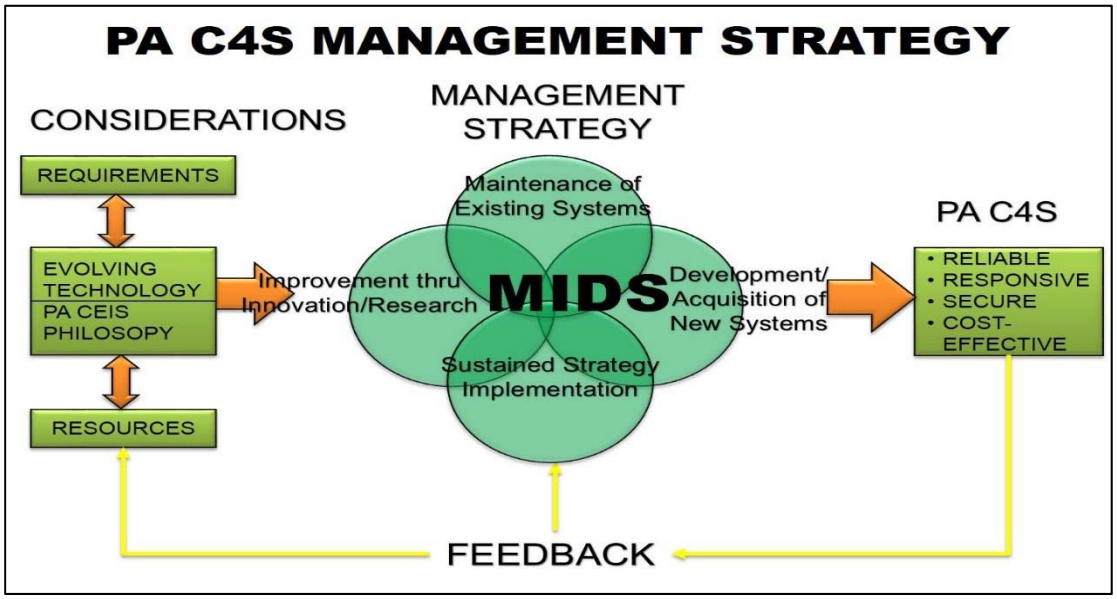


Figure 5.5 - PA C4S Management Strategy

0512. **Maintenance of Existing Systems.** The Philippine Army recognizes the importance of maintenance as follows, to extend the duration of every equipment beyond its service life.

a. **Preventive Maintenance –** This is emphasized at all levels of the organization, from the individual soldier to the biggest unit, involved in the operations and maintenance of C4S infrastructures.

b. **Restorative Maintenance –** Appropriate levels of maintenance shall be applied to the fullest, nevertheless, the capabilities of commercial companies to repair equipment shall also be tapped.

0513. **Improvement thru Innovations.** To provide immediate requirements of PA units, the principle of "appropriate technology" stipulates guidance in the pursuit of innovative solutions.

a. **Complementary Use of Available Commercial Systems –** These include practical use of cellular phones currently sold in the market as well as those commercially available handheld and base radios to complement communications requirements

b. **Self-Reliant Defense Posture (SRDP) Initiatives –** Some adaptations have to be done to suit the requirements when equipment and services available are not readily usable. Modifications are done to these equipment and services in order to maximize their intended purpose.

c. **System Enhancement by Leveraging Existing Facilities –** The liberalization of the telecommunications sector provided the AFP in general and the Philippine Army, in particular, the opportunity to leverage its existing telecommunications facilities in exchange for some services. Antenna tower and floor spaces, lot spaces, and other facilities can be offered to commercial telecommunications companies for their use. In return, cellular phone services, trunk lines, and transmission facility requirements may be received. This and other opportunities shall continuously be explored without sacrificing security.

0514. **Development of Appropriate Systems.** The Philippine Army continually develops appropriate systems to which, the AFP modernization program fund allocations are being used at present for the required capabilities development.

a. **Acquisition of New Equipment through the AFP Modernization Program. -** The PA Modernization fund component of the AFP modernization program has allowed the acquisition of two types of tactical radios, namely: the HF/SSB, 20-Watt, Manpack radio and the VHF/FM, 5-Watt, Handheld radio.

b. **Sustainable Capability Enhancement Program through Available Capital Outlay Fund –** Sustainable capability enhancements could only be pursued depending on the availability of capital outlay funds. This shall be vigorously pursued and justified in order to continuously develop capabilities that were not considered under the original AFP modernization program and which the usual maintenance and other operating expenses (MOOE) may not allow to be acquired.

c. **Sustainable Capability Enhancement Program by Leveraging Alliances –** The Philippine Army shall avail of the opportunity provided by the country's alliances to continually develop C4S capabilities.

0515. **Sustained Strategy Implementation –** The change in the chief-of-office affects the long-term implementation of an adopted strategy, thus, the following activities shall be conducted to ensure sustained quality result:

a. **Continuing Awareness on the Role of C4S Infrastructures -** Continuing C4S awareness and literacy program at all levels shall be addressed at all times. It is only through a fully aware and C4S literate army that a sustained implementation of the system is assured of.

b. **Continuing Review, Revision and Development of C4S Policies –** In the absence of concrete doctrine on C4S, the Philippine Army has continually developed policies, in the form of circulars and (SOP) standard operating system, to guide the management of C4S infrastructures. These policies shall be continuously reviewed and revised to keep them attuned to the change in the environment. Likewise, more policies shall be developed, as needed, to

continuously make the management of the C4S infrastructures of the organization relevant and worthwhile.

c. **Continuing Signal Corps Personnel Development –** In every dynamic organization, one of the most important resources that must be developed is the human resource. The Signal Corps of the Philippine Army is the human resource component, who ensures the overall effectiveness of the whole C4S infrastructure.

PA C4S Readiness and Sustainment

0516. In pursuit of the Army Transformation Roadmap (ATR) 2028, the Philippine Army's vision of *"Being a world-class Army that is a source of national pride"* shall involve the maximum utilization of available C4S equipment and facilities of the PA. The PA C4S Plans and Programs will supplement the attainment of ATR 2028 by sustaining the serviceability of on-hand C4S equipment and facilities, and enhancing/upgrading them through the projects under PA capability upgrade program (CUP), strategic initiatives of ATR 2028, and allocations from available capital outlay (CO) funds.

0517. However, the on-hand C4S equipment and existing facilities of PA are at risk of deterioration due to inconsistent implementation and non-observance of prescribed repair and maintenance procedures by PA units. In some instances, there were perennial reports on negligence, particularly on the part of the individual personnel who are operating these C4S equipment and facilities. This unfortunate reports on irresponsible practices led the facilities' eventual breakdown.

0518. Hence, the extension of the operational and service life of existing C4S equipment and facilities must be given equal importance by PA. This can be achieved through aggressive repair, rehabilitation and maintenance programs, appropriation of C4S maintenance and other operating expenses (MOOE) and the commitment of PA personnel to sustain the serviceability of C4S equipment and facilities in the PA inventory.

0519. This will provide appropriate tools for the command, control and administration of all PA units. Its sustained 24/7 operational status and maximum service life can be attained through the establishment of procedures for the conduct of preventive maintenance, scheduled maintenance, and repair by PA units, as well as, by the individual soldiers, who are responsible for its operation and use.

Training For Repair and Maintenance

0520. Repair and maintenance trainings for C4S equipment and facilities are part of the specialized C4S training responsibilities of ASR (P).

0521. TSS, ASR (P), in coordination with SIMBn, ASR (P), shall prepare all programs of instructions (POIs) for the various specialized C4S trainings of PA on its C4S equipment and facilities that include subjects/modules on repair and

maintenance. These POIs must be duly approved by commander, ASR (P) and submitted to this Headquarters (Attn: AC of S for C4S, G6) for notation.

0522. For synchronization of type of training/title, unit of primary responsibility (UPR) and manuals to be used, the following shall be observed:

Table 5.1 – Operations and Maintenance Training on C4S Equipment

LEVEL	TRAINING/TITLE	UPR	MANUAL
1	Operations and Maintenance Training	Signal Battalions of ASR(P) OPCON to the Infantry Divisions, and TSS, ASR(P)	Operator's Manual
2	Operations, Maintenance and Programming Training	TSS and SIMBn, ASR(P)	Operator's and Programming Manuals
3	Systems Operation, Maintenance and Programming Training	SIMBn, ASR(P) and concerned C4S Equipment Supplier	Systems Operation and Maintenance Manuals
4	Systems Operations and Repair Training	Concerned C4S Equipment Supplier	Systems Operations and Repair Manuals

0523. Only trained signal operators shall be allowed to operate and conduct scheduled and preventive maintenance procedures on C4S equipment and facilities of PA.

0524. All operations and maintenance training on C4S equipment and facilities must always emphasize the distinct procedures for scheduled and preventive maintenance, and a reminder that the operators are not allowed to perform experiments and other unauthorized operations, or connections, to the equipment.

0525. Signal career courses conducted by CSS, TRADOC, PA must incorporate in its POI the Level 1 (scheduled and preventive maintenance) principles and procedures on C4S equipment used by combat units, such as; tactical radios, telephone extender, public address system (PAS), geographical positioning system (GPS) and night fighting system (NFS).

0526. Appropriate training packages on newly acquired C4S equipment of PA must always be included in the contract/negotiation with the supplier. Moreover, technology transfer on its repair and maintenance must also be given emphasis so as to improve the level of competency of the C4S technicians of ASR (P) on the matter and extending the service equipment beyond warranty.

0527. The supplied operator's manual for C4S equipment must be included in the issuance directive and the acknowledge receipt of equipment (ARE) to ensure that the C4S Operators have a ready reference on its operation and preventive maintenance. Such operator's manual must be accounted at all times and considered as classified

document of the unit to prevent access by unauthorized personnel, to include the enemy of the state.

Programming of Funds

0528. Programming of MOOE for the repair and maintenance of Information and Communications Technology (ICT) equipment and facilities must be in accordance with the current command guidelines on the annual plan and budget (APB).

0529. HPA program directors shall program funds for the repair and maintenance of C4S equipment that are deployed to support their functional activities.

0530. PA Major Units shall program funds for maintenance of C4S equipment for their respective headquarters and subordinate units.

0531. Fund requirements for repair and maintenance beyond the capabilities of PA Major Units must be requested to this Headquarters (Attn: AC of S for C4S, G6) with the necessary data as basis for evaluation.

Delineation of Duties and Responsibilities

0532. The following shall be the standard repair and maintenance levels to be observed for common understanding of stakeholders and client units:

a. C4S equipment is limited to visual inspection of external parts; no use of repair kits, tools and test equipment, replacement of physical accessories/ancillaries, and; conduct of servicing with no involved spare parts. Other services may include physical connection of accessories/ancillaries and replacement of internal batteries. Operators must have completed an Operations and Maintenance Training in the use of the operator's manual only.

b. OG6, PA and Signal Battalions of Army Signal Regiment (Provisional), or ASR (P) and Signal Offices and Units of PA Major Subordinate Units, trained and authorized personnel from these Signal Offices and Units. C4S Facilities are limited to visual inspection of external parts, evaluation of physical relationship of parts; use of basic tools and test equipment to monitor voltage, amperes, continuity, etc. No use of repair kits, and conduct of servicing with no involved spare parts shall be allowed. Other services may include programming of appropriate software and assembly/disassembly of accessories/ancillaries. Operators must have completed trainings on operations, maintenance, and programming, and on the use of the operator's and programming manuals.

c. OG6, PA and Signal Battalions of ASR (P), and signal offices and units of PA major subordinate units trained and authorized technicians from these signal units and qualified technicians of the C4S equipment suppliers of PA. C4S equipment and C4S Facilities are to be repaired and maintained on depot level; all kinds of tools are allowed together with automated repair kits, special tools, and computer. Other activities may include; assembly/disassembly/replacement/ handling of the modules; testing of

individual electronic parts (transistors, capacitors, integrated circuit, resistors, diodes, etc.); and evaluation of circuits/boards. They also conduct systems reprogramming as prescribed by the manufacturer. Operators must be qualified technicians who completed a systems operations and repair training conducted by the concerned C4S equipment suppliers.

d. Same units/individuals in Level 3. Conduct of higher depot level repair and maintenance on C4S Equipment and facilities. Activities include those in Levels 1 to 3. They are allowed to use all kinds of tools, automated repair kits and special tools and computer – aided test equipment (special equipment dedicated/proprietary to the brand/device). To be qualified technicians, they must have completed systems operations and repair training conducted by the concerned C4S equipment suppliers. As to the use of the Systems Operations and Repair Manuals, other activities may include assembly/disassembly/replacement/ handling of the modules, testing of individual electronic parts (transistors, capacitors, integrated circuit, resistors, diodes, etc.) and evaluation of circuits/boards. Systems reprogramming may also be conducted as prescribed by the manufacturer.

Accountability and Responsibility

0533. The following shall provide guidance for the stewardship of C4S facilities. It describes information management-related duties for individuals, as well as mechanisms for monitoring the performance of those responsibilities.

a. C4S equipment and C4S facilities must be duly received, through an acknowledgement receipt of equipment (ARE), by recipient/using units, or end users, duly signed by the unit Property Officer (UPO), or the responsible supply officer (RSO).

b. All issuances of C4S equipment and C4S facilities must include a detailed inventory of its component parts (with or without serial number), accessories and ancillaries to be included as an accountability of the UPOs and RSOs.

c. The inventory of recipient/using units on C4S equipment and facilities must include the accessories and ancillaries so as to consider its serviceability during the conduct of technical inspection.

d. Unaccounted parts, accessories and ancillaries of C4S equipment and facilities must be reflected whenever there is a transfer of property responsibility.

e. C4S equipment and facilities that are declared as beyond economical repair (BER) must be turned-over to the nearest FSSU of ASCOM for proper disposal. A copy of the property turn-in slip (PTIS) must be kept on record by the using unit.

f. Lost, stolen, and unauthorized use of C4S equipment and facilities must immediately be reported to this headquarters (Attn: AC of S for C4S, G6), through a spot report, within three (3) days of the incident for subsequent information to higher headquarters. Such report must include details on the

specific equipment/facility, serial number(s), name of user, using unit and circumstances on the lost, theft and unauthorized use.

g. An appropriate investigation must be conducted by the concerned unit on the reported loss, theft and unauthorized use of the C4S equipment and facilities, Subsequently Investigation Report must be submitted to this Headquarters (Attn: AC of S for C4S, G6) NLT 15 days from the date of incident. In the event of loss or theft, a report of survey must be forwarded to the Headquarters of the concerned PA Major Subordinate Unit for the dropping of accountability on the C4S equipment.

h. C4S equipment and facilities that were found to be unserviceable/defective due to personnel negligence during the conduct of technical inspection by technicians of ASR(P), and in coordination with the concerned C4S equipment suppliers of the equipment/facility, must be reported to Commander, ASR(P) of this headquarters (Attn: AC of S for C4S, G6).

i. The report of commander, ASR (P) shall be used as reference in the conduct of an investigation to the End User/recipient unit on the circumstances that led to the unserviceable/defective C4S equipment and C4S facilities due to personnel negligence.

Physical Protection and Security

0534. The following shall serve as procedures to ensure that C4S facilities are well-maintained to prevent damage, loss, or misuse.

a. Serviceable and unserviceable C4S equipment must always be stored in a secured place at the unit headquarters/detachments with appropriate exterior/interior doors and locks, and C4S facilities must always be accorded with physical security measures prescribed in existing AFP/PA regulations.

b. During combat operations, C4S equipment must always be placed in the issued carrying case(s).

c. When used outside camps/detachments, C4S equipment must not be left unattended/unsecured at all times.

d. C4S equipment must always be housed in a building/housing that has protection from the environment and weather extremities, such as; sun, rain, typhoon, etc.

e. C4S facilities must have appropriate firefighting devices and measures such as; fire extinguishers with Category C (electrical equipment) protection, evacuation priority markers – priority 1 for C4S equipment, fire evacuation drills, and others to protect the equipment thereat.

f. Unserviceable C4S equipment must also be accorded with due security at all times, especially when transported from one place to another for repair, turn-in or for other purposes.

Transportation

0535. The following are guidelines in order to ensure that C4S facilities are always kept in a secured location.

a. C4S equipment sent for repair must always be transported in the most secured means possible.

b. C4S equipment must be placed in a secure portion of the transportation facility at all times.

c. Appropriate security measures must be employed by the courier/escort of C4S equipment whenever it is transmitted through commercial means of sea, air, and land transports.

Acquisition and Disposal of C4S Equipment

0536. The following are institutionalized standards for acquisition or disposal of C4S equipment to ensure that equipment go through appropriate processes of evaluation and approval, for implementation of proper information disclosure.

Demilitarization/Disposal

0537. All C4S equipment declared beyond economical repair (BER) shall be turned-in to FSSUs of ASCOM and accompanied with the appropriate technical inspection report (TIR).

0538. No spare parts, accessories and ancillaries of C4S equipment must be disposed by the using unit without proper authorization.

0539. For safety reasons on potential hazard to humans and the environment, only ASCOM shall dispose turned-in C4S equipment, spare parts, accessories and ancillaries that are declared BER.

Procedures

0540. C4S operators must conduct regular preventive maintenance procedures on issued C4S equipment and C4S facilities.

0541. In the event of an equipment breakdown, or facility failure, the following guidelines apply:

a. Conduct technical tests procedures, as prescribed in user level training for possible restoration of the C4S equipment and C4S facilities, with the supervision of the designated signal officer.

b. Upon confirmation of the equipment breakdown and facility failure, individuals should send the involved equipment to the concerned signal battalion of ASR (P) placed OPCON to the infantry divisions, for technical

inspection, Otherwise, technical inspection may be conducted as to size, quantity of the equipment, etc., on-site upon the request of the concerned unit to the division G6.

c. The signal battalion personnel can conduct technical inspection and evaluation on the unserviceable C4S equipment and facilities.

d. In order to provide a systematic guide on the steps to be performed by the concerned signal units and C4S equipment suppliers for the repair of unserviceable C4S equipment, the repair flow charts shall be observed; 1) unserviceable equipment handled at the level of PA major subordinate units, 2) unserviceable equipment handled at the level of headquarters, ASR (P), 3) unserviceable equipment handled at the level of SIMBn, ASR (P), and 4) unserviceable equipment handled at the level of C4S equipment suppliers.

0542. Repair of C4S equipment still covered by service warranty and beyond the capability of SIMBn, ASR (P) shall be sent to the concerned C4S Equipment Supplier to be processed in accordance with the flow chart for said supplier.

0543. Batteries of C4S equipment must be removed whenever it will be left for safekeeping at the designated storage area. If necessary, these batteries must be included in the packaging of unserviceable equipment for verification of its status and the possible cause behind its breakdown.

0544. No experiments, or use for non-intended purpose, must be performed on C4S equipment. In the event that there is a need to explore the use of C4S equipment, beyond the prescribed utilization, it must be conducted only by trained technicians of ASR (P).

0545. All C4S equipment must always be placed inside the issued carrying case when not in use. Unserviceable C4S equipment sent for repair must be enclosed in the issued carrying case with all the required accessories/ancillaries, as prescribed/required by ASR (P). Concerned units of ASR (P) must not receive C4S equipment sent for repair not placed inside the issued carrying case and the required accessories/ancillaries.

0546. HPA may front load and preposition priority C4S equipment, accessories/ancillaries, spare parts and other supplies to the appropriate unit of PA unit, subject to its availability and on need case basis. These include items that are allocated for “Direct Exchange” to field units, subject to the existing logistical procedures.

0547. ASR (P) shall establish a database for the repair of C4S equipment with data on, but not limited to: serial number, date issued, using unit, repair conducted, accessories/ancillaries and spare parts replaced, among others. This database shall serve as reference for future repairs of said equipment.

0548. ASR (P) shall update the recipient/using units on the status of their respective C4S equipment sent for repair, through the headquarters of the PA major subordinate units.

Glossary

AFP Modernization Program
The activities undertaken by the AFP to develop its capabilities as mandated by RA 10349 or the Implementing Guidelines, Rules and Regulations of the Revised AFP Modernization Act

Architecture
A framework or structure that portrays the relationships between and among elements of the subject force, system or activity.

Asymmetric Warfare
The use of conventional or unconventional weapons or approaches to war fighting to nullify an opponent's conventional advantage.

C4S Philosophy
Set of principles, concepts and convictions guiding the management of C4S infrastructure.

Command
It is the authority that a commander in the military service lawfully exercises over subordinates, by virtue of rank or assignment.

Command and Control
The exercise of authority and direction by a properly designated commander over assigned and attached forces in the accomplishment of the mission. Command and Control functions are performed through an arrangement of personnel, equipment, communications, facilities and procedures employed by a commander in planning, directing, coordinating and controlling forces and operations in the accomplishment of the mission.

Command and Control Communications System
The organization of resources that the commander uses to help him plan, direct, coordinate and control military operations to ensure mission accomplishment.

Communications
A method or means of conveying information of any kind from one person or place to another.

Computer
An electronic device that receives input data, puts them into storage, operates on them according to a program, and outputs the result to the users.

Economy of Force
Employing the least possible resources elsewhere other than at the decisive time and place.

HF/SSB
High Frequency/Single Side Band (3-30 Mhz), frequency and transmission method of specification for radio sets used for long distance ground communications.

Information
Refers to processed data in meaningful form.

Information System
A computer-based sourcing, manipulating and storing data for the purpose of generating information for the use of organizations.

Information Technology
Refers to computers, telecommunications and office systems technologies.

Infrastructure
The basic facilities, equipment, and installation needed for the function of a system, network, or integrated network.

Joint Operations
Combat operations that involve the employment of forces from two or more services (PA, PN and PAF).

Knowledge-based system
It is a collection of facts, rules, and procedures which are organized into schemas.

Mission Command
The shared understanding between commanders and his subordinates of the environment and its expectations which enable every member to exercise disciplined initiative.

National Command Authority
The national government structure that exercise control over all armed forces.

National Military Strategy
The document that sets the AFP focus on current and emerging security concerns and adjusts the AFP strategy based on more realistic conditions. It set the national military objectives, the elements of strategy and the force required.

Radio Net
It is composed of two or more radios operating in the same frequency for the specific purpose of interchanging information.

Signal Corps
The service branch of the Philippine Army primarily tasked in the management and operations of army C4S.

UHF/FM
Ultra High Frequency/Frequency Modulation (403-433 Mhz), frequency and modulation specification of radio sets for internal communications.

VHF/FM
Very High Frequency/Frequency Modulation (30-80 Mhz), frequency and modulation specification of radio sets for short distance communications.

Abbreviation and Acronym List

AFP FCS	AFP Fixed Communications System
AFPMP	AFP Modernization Program
AFPOS	AFP Occupational Specialty
BSO	Brigade/Battalion Signal Officer
C-E	Communications-Electronics
C4S	Command and Control Communications, and Cyber Systems
C2	Command and Control
CMO	Civil Military Operations
COMCAM	Combat Camera
COTS	Commercially Off the Shelf
DATACOM	Data Communications
DSO	Division Signal Officer
EEZ	Exclusive Economic Zone
EW	Electronic Warfare
G6	Assistant Chief of Staff for C4S, G6, PA
HF/SSB	High Frequency/Single Side Band (3-30MHz)
LAN	Local Area Network
LIS	Logistics Information System
IS	Information Systems
ISO	Internal Security Operations, Information Systems Office/Officer
MIS	Management Information Systems
NBC	Nuclear Biological Warfare
NCA	National Command Authority
PANET	Philippine Army Intranet
PAWSSU	PA Wide Support and Separate Units
PIS	Personnel Information System
RDF	Rapid Deployment Force
SC	Signal Corps
SRDP	Self-Reliant Defense Posture
UHF/FM	Ultra High Frequency/Frequency Modulation (403-433 MHz)
VHF/FM	Ultra High Frequency/Frequency Modulation (30-80 Mhz)

VI	Visual Information
VSAT	Very Small Aperture Terminal